

For topic 10, I suggest referring to the Organic Chemistry AS pdf instead as it contains both from unit 1 and unit 2 which helps you connect all ideas and learn better. Nevertheless, the content here is almost same just a bit expanded.

10A — General Principles of Organic Chemistry

10.1 — Classification of Organic Reactions

You are expected to **identify and classify** the following reaction types, including those introduced in Unit 1:

Reaction Type	Definition	Example
Addition	Two molecules combine to form one product (usually involving C=C)	$\text{CH}_2=\text{CH}_2 + \text{Br}_2 \rightarrow \text{CH}_2\text{Br}-\text{CH}_2\text{Br}$
Elimination	A small molecule (e.g., H_2O or HX) is removed from a larger molecule	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3 \rightarrow \text{CH}_2=\text{CHCH}_3 + \text{H}_2\text{O}$
Substitution	An atom/group in a molecule is replaced by another	$\text{CH}_3\text{CH}_2\text{Br} + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{Br}^-$
Hydrolysis	Splitting of a molecule by reaction with water or OH^-	$\text{CH}_3\text{CH}_2\text{Cl} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{HCl}$
Oxidation	Loss of electrons, hydrogen, or gain of oxygen (mainly for alcohols)	$\text{CH}_3\text{CH}_2\text{OH} + [\text{O}] \rightarrow \text{CH}_3\text{CHO} + \text{H}_2\text{O}$
Reduction	Gain of electrons, hydrogen, or loss of oxygen	$\text{CH}_3\text{COOH} + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{H}_2\text{O}$
Polymerisation	Many small molecules (monomers) join to form a long chain (polymer)	$n\text{CH}_2=\text{CH}_2 \rightarrow -[\text{CH}_2-\text{CH}_2]-_n$

10.2 — What Is a Reaction Mechanism?

A **reaction mechanism** is a detailed, step-by-step description of how bonds are broken and formed during a reaction.

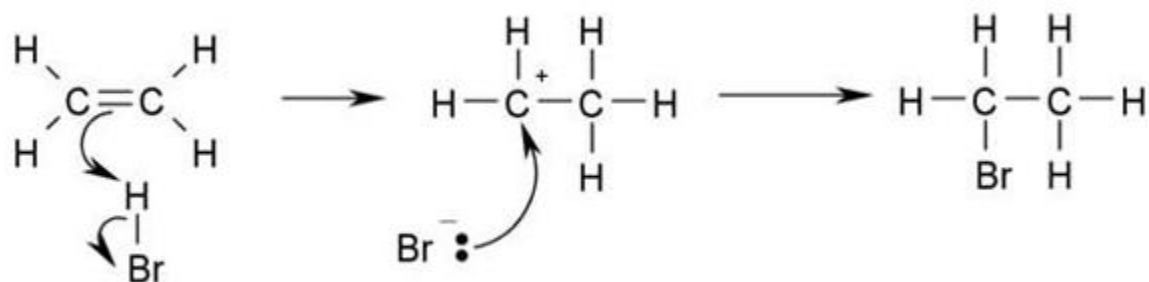
It shows:

- Movement of **electron pairs** using curly arrows ($\curvearrowright$)
- Formation of **intermediates** or **transition states**
- Overall conversion from reactants to products

Mechanisms you'll encounter:

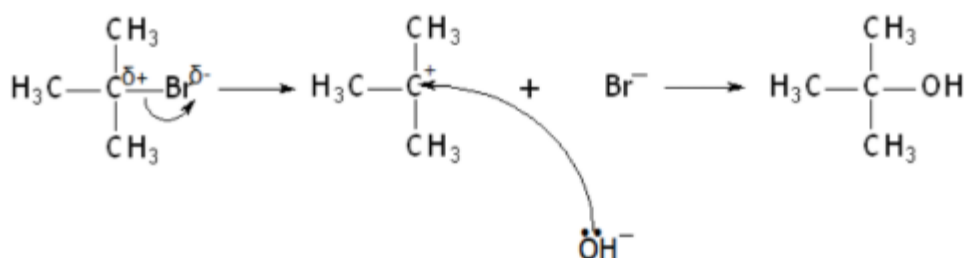
- **Nucleophilic substitution** (S_N) — Halogenoalkanes
- **Elimination** — Alcohols or halogenoalkanes

- **Electrophilic addition** — Alkenes



Electrophilic
addition

Example of mechanism



Nucleophilic
substitution

10.3 — Heterolytic vs Homolytic Fission

Type of Bond Breaking	Definition	Products Formed
Heterolytic fission	Bond breaks unevenly , both electrons go to one atom	Ions (e.g., Cl^- , CH_3^+)
Homolytic fission	Bond breaks evenly , each atom takes one electron	Radicals (e.g., $\text{Cl}\cdot$)

Heterolytic fission is most common in **nucleophilic and electrophilic reactions**, where **charged species** are involved.

10.4 — What Is a Nucleophile?

A **nucleophile** is a species that **donates a lone pair of electrons** to form a new covalent bond.

Nucleophiles are:

- **Electron-rich**

- Often **negatively charged** or have δ^- regions
- Attracted to **electron-deficient (δ^+) centres**

Nucleophile	Structure	Function
OH^-	Hydroxide	Attacks δ^+ carbon in halogenoalkanes
NH_3	Ammonia	Donates lone pair on N
CN^-	Cyanide	Extends carbon chain
H_2O	Water	Weak nucleophile

10.5 — Bond Polarity and Mechanism Type

- A **polar bond** (e.g., C–Br) has δ^+ on carbon and δ^- on halogen
- The δ^+ carbon becomes the **target for nucleophiles**
- This leads to **nucleophilic substitution reactions** in halogenoalkanes

If the bond is **non-polar**, **homolytic fission** is more likely → radical mechanisms (e.g., in photochemical reactions)

10.6 – Naming halogenoalkanes (refer to organic chemistry pdf instead)

10.7 — Primary, Secondary, and Tertiary Halogenoalkanes

A. Why Classification Matters

The structure of the halogenoalkane affects:

- **How easily the C–X bond breaks**
- **The reaction mechanism**
- **The reactivity and type of product formed**

B. The Classification System

Type	Definition	General Structure
Primary (1°)	The carbon attached to the halogen is bonded to only one other carbon	$\text{R}-\text{CH}_2-\text{X}$
Secondary (2°)	The carbon attached to the halogen is bonded to two other carbons	$\text{R}-\text{CH}(\text{X})-\text{R}'$
Tertiary (3°)	The carbon attached to the halogen is bonded to three other carbons	$\text{R}_3\text{C}-\text{X}$

C. Examples

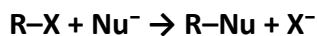
Compound	Type	Explanation
$\text{CH}_3\text{CH}_2\text{Cl}$	Primary	Halogen on CH_2 , bonded to one other C
$\text{CH}_3\text{CHClCH}_3$	Secondary	Halogen on central C, bonded to 2 other Cs
$(\text{CH}_3)_3\text{CCl}$ (tert-butyl chloride)	Tertiary	Halogen on C bonded to 3 methyl groups

10.8 — Reactions of Halogenoalkanes

A. Nucleophilic Substitution Reactions

In these reactions, the halogen atom (a leaving group) is **replaced** by a **nucleophile**.

General Reaction:



Where:

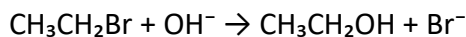
- $\text{R}-\text{X}$ = halogenoalkane
- Nu^- = nucleophile
- X^- = halide ion

10.8(i) — Reaction with Aqueous Potassium Hydroxide (KOH)

- **Conditions:** Heat under **reflux** in **aqueous solution**
- **Nucleophile:** OH^- (from KOH)

- **Type: Nucleophilic substitution**
- **Product: Alcohol**

Example:



(*Bromoethane* $\rightarrow$ *Ethanol*)

10.8(ii) — Reaction with Ethanolic Potassium Hydroxide

- **Conditions:** Heat under **reflux** in **ethanol**
- **OH⁻** now acts as a **base**, not a nucleophile
- **Type: Elimination** (see below)
- **Product: Alkene + H₂O + X⁻**

Example:



(*2-Bromopropane* $\rightarrow$ *Propene*)

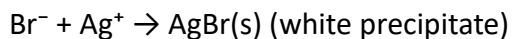
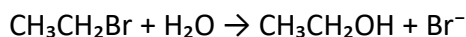
More on this under **elimination** reactions later in this message.

10.8(iii) — Reaction with Silver Nitrate in Ethanol

Used to study **rate of hydrolysis**. Water acts as the nucleophile.

- **Purpose:** To observe rate of halogenoalkane hydrolysis
- **Ethanol** used as solvent to dissolve halogenoalkane
- **Water** acts as nucleophile, replacing halide with OH

Reaction:



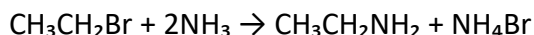
- Time taken for precipitate = measure of rate
- **I⁻** forms fastest $\rightarrow$ **strongest reducing agent**
- **Cl⁻** forms slowest $\rightarrow$ **strongest C–X bond**

10.8(iv) — Reaction with Ammonia (NH₃)

- **Conditions:** Heat with **excess concentrated NH₃** in **ethanol** under pressure
- **Nucleophile:** NH₃ (neutral molecule)

- **Type:** Nucleophilic substitution
- **Product:** Primary amine (R-NH_2)

Example:

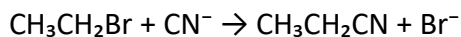


- Excess NH_3 ensures **monosubstitution**
- A salt (e.g., NH_4Br) is also formed

10.8(v) — Reaction with Potassium Cyanide (KCN)

- **Conditions:** Reflux with **ethanolic KCN**
- **Nucleophile:** CN^-
- **Type:** Nucleophilic substitution
- **Product:** Nitrile ($\text{R-C}\equiv\text{N}$) $\rightarrow$ used for **carbon chain extension**

Example:



(*Bromoethane $\rightarrow$ Propanenitrile*)

This is a **useful synthetic step** in organic chemistry.

B. Elimination Reactions (with OH^- in ethanol)

OH^- ion acts as a **base** and removes a proton (H^+) from a carbon **adjacent to the one bonded to the halogen**, forming an **alkene**.

Conditions:

- **Hot ethanolic KOH**
- **Reflux**

General Reaction:



Example:



(*2-Bromopropane $\rightarrow$ Propene*)

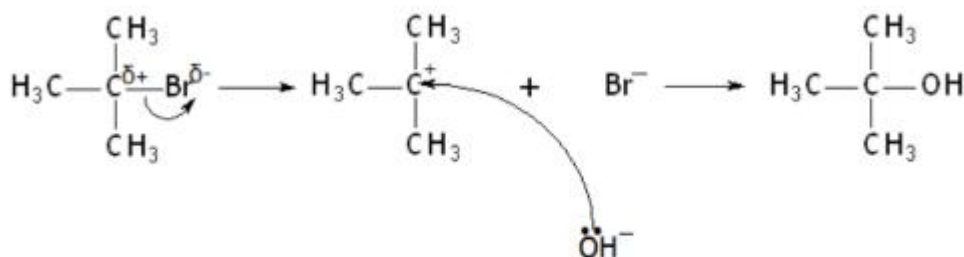
C. Competing Reactions: Substitution vs Elimination

Conditions	Predominant Reaction	Product Type
Aqueous KOH, warm	Substitution	Alcohol
Ethanolic KOH, hot	Elimination	Alkene

10.9 — Mechanism of Nucleophilic Substitution in Primary Halogenoalkanes

Refer to the organic chemistry pdf, this is just drawing the mechanism only, nothing else

Example of mechanism



Step 1: Formation of Carbocation

- The **C–Br bond breaks heterolytically**.
- The **bromine atom takes both electrons**, forming a Br^- ion.
- A **tertiary carbocation** is formed at the central carbon.

Curly Arrow:

- From the **C–Br bond** to **Br**, showing departure of the halide ion.

Intermediate Formed:

- $(\text{CH}_3)_3\text{C}^+$ (stable tertiary carbocation)
- Br^- (leaving group)

Step 2: Nucleophilic Attack by OH^- (Fast Step)

- The lone pair on the **hydroxide ion (OH^-)** attacks the **carbocation $(\text{CH}_3)_3\text{C}^+$** .
- A **new covalent bond** forms between the central carbon and the oxygen atom of OH^- .

Curly Arrow:

- From the **lone pair on OH^-** to the **positively charged carbon**.

Product Formed:

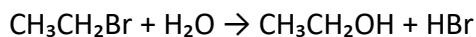
- $(\text{CH}_3)_3\text{COH}$ (tert-butanol)
- Br^- (remains as spectator ion)

10.10 — Comparing Rates of Hydrolysis of Halogenoalkanes

A. What Is Hydrolysis?

Hydrolysis of halogenoalkanes is the reaction with **water** (or aqueous hydroxide) that **replaces the halogen atom with an $-\text{OH}$ group**, forming an **alcohol**.

Example:



B. How Do We Compare Rates?

We observe **how quickly the halide ion is released**, using **$\text{AgNO}_3(\text{aq})$** to form a **precipitate** with the halide ion.

Set-up:

- Use **ethanol** to dissolve halogenoalkanes (they are non-polar)
- Add **aqueous silver nitrate (AgNO_3)**
- Time how long it takes for a **precipitate** to appear

Observation = Precipitate formation

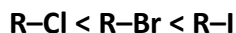
Faster precipitate = Faster hydrolysis = Weaker C-X bond

C. Silver Nitrate Test for Halide Ions

Halogenoalkane	Halide Ion Released	Silver Halide Formed	Precipitate Colour
Chloroalkane (R-Cl)	Cl^-	AgCl	White
Bromoalkane (R-Br)	Br^-	AgBr	Cream
Iodoalkane (R-I)	I^-	AgI	Yellow

D. Trend in Rate of Hydrolysis

Rate of hydrolysis increases as you go down the halogen group:



Explanation:

- C-I bond is **longer and weaker** $\rightarrow$ breaks more easily
- C-Cl bond is **shorter and stronger** $\rightarrow$ more energy needed to break

E. Experimental Controls and Validity

To ensure a **fair test** when comparing hydrolysis rates:

Factor to Keep Constant	Why?
Concentration of halogenoalkanes	Affects collision frequency
Volume and concentration of AgNO_3	Affects visibility and rate of ppt formation
Temperature	Affects reaction rate
Solvent (ethanol + water)	Ensures uniform solubility
Same structure (e.g. all primary halogenoalkanes)	Controls for SN1/SN2 variation

F. Summary of Rate Trend

“Rate of hydrolysis increases from chloroalkane to iodoalkane because the **C–X bond strength decreases**, so **less energy is needed** to break the bond and form the carbocation or transition state.”

10.11 is core practical so skipped

10.12 — Reactivity Trend of Primary, Secondary, and Tertiary Halogenoalkanes

A. What Affects Reactivity?

The reactivity of halogenoalkanes depends on:

1. **Structure** (primary, secondary, tertiary)
2. **Mechanism** involved (SN1 vs SN2)
3. **Carbocation stability**
4. **Steric hindrance**
5. **C–X bond strength** (covered in 10.10)

B. Reactivity Order (Based on Structure)

Tertiary > Secondary > Primary

This order is observed in **aqueous hydrolysis reactions** using water or OH^- as nucleophiles.

C. Explanation

Type	Reason for Reactivity
Tertiary	Forms a stable tertiary carbocation
Secondary	Intermediate stability and reactivity
Primary	No stable carbocation

D. Why Tertiary Halogenoalkanes React Fastest

- **Tertiary carbocations** are most stable
- So tertiary halogenoalkanes lose the halide quickly, speeding up the reaction.

E. Example:

Hydrolysis with water:

$(\text{CH}_3)_3\text{C}-\text{Br}$ (tertiary) $\rightarrow$ reacts fastest

$\text{CH}_3\text{CHBrCH}_3$ (secondary) $\rightarrow$ medium

$\text{CH}_3\text{CH}_2\text{Br}$ (primary) $\rightarrow$ slowest

10.13 — Bond Enthalpy and Reactivity of Halogenoalkanes (Cl, Br, I)

A. Key Idea

The **reactivity of halogenoalkanes increases** from **chloro-** $\rightarrow$ **bromo-** $\rightarrow$ **iodoalkanes**

This is due to **decreasing bond enthalpy** of the **carbon–halogen bond (C–X)**.

B. Bond Enthalpies (kJ mol^{-1})

Bond	Approx Bond Enthalpy	Strength
C–Cl	$\sim 340 \text{ kJ mol}^{-1}$	Strongest
C–Br	$\sim 280 \text{ kJ mol}^{-1}$	Medium
C–I	$\sim 240 \text{ kJ mol}^{-1}$	Weakest

C. Why It Matters

- In **nucleophilic substitution**, the **C–X bond must break**
- The **weaker** the bond, the **easier** and **faster** the reaction
- Therefore:
 - **Iodoalkanes react fastest**

- Chloroalkanes react slowest

D. Bond Polarity ≠ Reactivity

Although **C–Cl bonds are more polar** than C–Br or C–I:

C–X Polarity	$\delta^+C-\delta^-X$ Trend	Would suggest...
C–Cl > C–Br > C–I	Cl is most polar	Cl [–] should be easiest to leave

(note: this refers to bond polarity not the ion being polarized)

BUT bond strength matters **more than polarity**

Actual reactivity trend:

Cl < Br < I

Reason: Breaking the C–X bond is the rate-determining step, and this depends on bond enthalpy.

E. Experimental Confirmation (from 10.10)

Hydrolysis with water + AgNO₃ shows:

Halogenoalkane	Precipitate Formation Time	Reactivity
R–Cl	Slow	Low
R–Br	Medium	Moderate
R–I	Fast	High

10.14 – core practical

10.15 – naming alcohols => refer to the organic chemistry pdf

10.16 — Primary, Secondary, and Tertiary Alcohols

A. Classification Based on the Carbon Bearing the –OH Group

The classification depends on how many carbon atoms are bonded to the carbon that carries the –OH group.

Type	Structure (R = alkyl group)	Defining Feature
Primary (1°)	$\text{R-CH}_2\text{-OH}$	–OH on carbon attached to 1 other carbon
Secondary (2°)	R-CH(OH)-R'	–OH on carbon attached to 2 other carbons
Tertiary (3°)	R-C(OH)(R')(R'')	–OH on carbon attached to 3 other carbons

B. Structural Examples

Name	Structure	Type
Ethanol	$\text{CH}_3\text{CH}_2\text{OH}$	Primary
Propan-2-ol	$\text{CH}_3\text{CH(OH)CH}_3$	Secondary
2-Methylpropan-2-ol	$(\text{CH}_3)_3\text{COH}$	Tertiary

10.17 — Reactions of Alcohols and 10.18 oxidation of alcohol

Reaction Type	Reagent(s) & Conditions	Example Equation	Mechanism / Notes	Color Change
Combustion	O ₂ (ignition)	$\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$	Complete or incomplete combustion	Pale blue flame
Reaction with Sodium	Na metal (dry alcohol)	$\text{C}_2\text{H}_5\text{OH} + \text{Na} \rightarrow \text{C}_2\text{H}_5\text{ONa} + \frac{1}{2}\text{H}_2 \uparrow$	H ₂ gas released (effervescence)	—
Dehydration (Elimination)	Conc. H ₂ SO ₄ / H ₃ PO ₄ , heat (170°C)	$\text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$	Forms alkene	—
Oxidation (1° alcohol)	Acidified K ₂ Cr ₂ O ₇ Distillation → Aldehyde Reflux → Carboxylic acid	$\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{CHO} \rightarrow \text{CH}_3\text{COOH}$	Orange → Green	
Oxidation (2° alcohol)	Acidified K ₂ Cr ₂ O ₇ , reflux	$\text{CH}_3\text{CHOHCH}_3 \rightarrow \text{CH}_3\text{COCH}_3$	Forms ketone	Orange → Green
Oxidation (3° alcohol)	Acidified K ₂ Cr ₂ O ₇	No reaction	No H on C–OH carbon	Stays orange
Reaction with PCl₅ (chlorination)	PCl ₅ (dry, room temp)	$\text{ROH} + \text{PCl}_5 \rightarrow \text{RCl} + \text{POCl}_3 + \text{HCl}$	Steamy white fumes (HCl)	—
Reaction with PI₃(iodination)	PI ₃ made in situ from red phosphorus + I ₂	$3\text{ROH} + \text{PI}_3 \rightarrow 3\text{RI} + \text{H}_3\text{PO}_3$	Nucleophilic substitution	—
Reaction with PBr₃	PBr ₃ (or made in situ)	$3\text{ROH} + \text{PBr}_3 \rightarrow 3\text{RBr} + \text{H}_3\text{PO}_3$	Forms bromoalkanes	—
Reaction with HBr(bromination)	50% conc. H ₂ SO ₄ + NaBr, heat under reflux	$\text{ROH} + \text{HBr} \rightarrow \text{RBr} + \text{H}_2\text{O}$	HBr made in situ , S _N reaction	—

Note: Oxidation of alcohols using KMnO₄ results in the same products as oxidation with K₂Cr₂O₇. Only difference is that the color changes from purple to colorless

10.19 — Preparation and Purification of Organic Liquids

A. Heating Under Reflux

Used to **heat a reaction mixture for a long time** without losing volatile substances.

Apparatus:

- Round-bottom flask
- **Vertical condenser**
- Heat source (e.g. water bath or heating mantle)

Function:

- Vapours condense and return to the flask → **prevents loss of product or reagents**
- Common in oxidising alcohols (e.g. primary alcohol → carboxylic acid)

B. Distillation

Used to **separate a liquid product** from a reaction mixture based on **boiling point**.

Apparatus:

- Round-bottom flask
- Thermometer
- Condenser (angled)
- Collection flask

Function:

- Heat mixture until desired product distils at its boiling point
- **Used to isolate aldehydes** (e.g. ethanol → ethanal)

C. Separation Using a Separating Funnel

Used to **separate immiscible liquids** (usually organic product and aqueous layer)

Steps:

1. Transfer mixture to separating funnel
2. Allow layers to settle (denser liquid sinks)
3. Remove the **aqueous layer** (usually the bottom layer)
4. Retain the **organic layer** (contains product)

Use:

- After reactions like **halogenoalkane prep** or **organic extraction**

D. Drying the Organic Product

Used to **remove traces of water** from the organic layer

Drying agents:

- **Anhydrous MgSO_4 or CaCl_2**
- Add until the liquid goes from **cloudy to clear**

Procedure:

1. Add drying agent to organic layer
2. Swirl and leave for 5–10 minutes
3. Filter off the solid

E. Redistillation

Re-distil the product to purify it by collecting the **fraction at the correct boiling point**

Purpose:

- Remove **impurities** with different boiling points
- Increases **purity and identity confidence**

F. Boiling Point Determination

Used to check the **identity and purity** of a product

- Pure compounds have **sharp, defined boiling points**
- Impure compounds show **broader boiling ranges**

Technique	Purpose
Reflux	Continuous heating without product loss
Distillation	Separation based on boiling point
Separating funnel	Separation of immiscible layers
Drying	Remove water using anhydrous salt
Redistillation	Further purification
Boiling point	Check identity/purity

10.21 — Mass Spectrometry (MS)

A. What Is Mass Spectrometry?

Mass spectrometry is an analytical technique used to determine the **relative molecular mass (Mr)** and **structure of organic compounds** by measuring the **mass-to-charge ratio (m/z)** of ionised fragments.

B. Key Features of a Mass Spectrum

A mass spectrum displays:

- **x-axis** = **m/z** (mass-to-charge ratio)
- **y-axis** = **Relative abundance**

C. Key Peaks to Identify

Peak Type	What It Tells You
Molecular ion peak (M^+)	m/z value = Mr of the molecule
Base peak	Tallest peak = most stable fragment
Fragment peaks	Arise from breakdown of molecule

D. Example: Mass Spectrum of Butan-2-one ($CH_3COCH_2CH_3$)

- **Molecular ion peak** at **m/z = 72**
- Common fragments:
 - CH_3CO^+ at m/z = 43
 - CH_3^+ at m/z = 15
- **Base peak** = m/z 43

E. Interpreting Fragmentation Patterns

Fragments help deduce **which groups are present** based on **typical bond breakage patterns**

Base fragment: The **base fragment** is the ion responsible for the **tallest peak in a mass spectrum**, representing the **most stable and most abundant ion** formed during fragmentation

Common Fragments:

Fragment Ion	m/z	Often Seen In...
CH_3^+	15	Alkanes, alcohols
OH^+	17	Alcohols
H_2O^+	18	Alcohols, carboxylic acids
CO^+	28	Aldehydes, ketones
CH_3CH_2^+	29	Alkanes
$\text{C}=\text{O}^+$	28	Aldehydes, ketones
CH_3CO^+	43	Ketones, esters

Most Common Base Fragment Ions

Ion Name	Formula	m/z	Structure/Group	Common Sources	Why It's Stable
Methyl cation	CH_3^+	15	$-\text{CH}_3$	Alkanes, alcohols	Small, low-energy carbocation
Ethyl cation	C_2H_5^+	29	$-\text{CH}_2\text{CH}_3$	Alkanes, esters	Slightly more stable primary carbocation
Propyl cation	C_3H_7^+	43	$-\text{CH}_2\text{CH}_2\text{CH}_3$ or CH_3CHCH_3	Alkanes	Moderately stable primary carbocation
Acetyl cation	CH_3CO^+ (RCO^+)	43	$-\text{COCH}_3$	Ketones, aldehydes, esters	(reason not required)
Hydroxymethyl	CH_2OH^+	31	$-\text{CH}_2\text{OH}$	Primary alcohols	(reason not required)
Water (neutral)	H_2O^+	18	$\text{H}-\text{O}-\text{H}$	Alcohols, carboxylic acids	Stable neutral molecule

10.22 — Infrared Spectroscopy (IR)

A. What Is IR Spectroscopy?

IR spectroscopy involves passing **infrared light** through a compound. **Covalent bonds absorb specific IR frequencies** and begin to **vibrate (stretch or bend)**.

- Each type of bond absorbs a **unique range** of IR radiation
- The absorption is recorded on a **spectrum** showing **transmittance vs wavenumber (cm^{-1})**

B. Axes of an IR Spectrum

Axis	What It Represents
X-axis	Wavenumber (cm^{-1}), inverse of wavelength
Y-axis	% Transmittance (how much IR passes through)
Peaks = Absorption	Lower % transmittance = stronger absorption (bond absorbs IR)

C. Interpreting an IR Spectrum

You identify **functional groups** by matching the **absorption peaks** (also called "troughs") to **characteristic wavenumber ranges**.

Key Region to Focus:

- 1500–4000 cm^{-1} = **Functional group region**
- < 1500 cm^{-1} = **Fingerprint region** (unique to each compound, not used for general ID)

D. Important Functional Group Absorptions

Bond / Group	Wavenumber Range (cm^{-1})	Description
O–H (alcohol)	3200–3550, broad	Broad, smooth absorption
O–H (carboxylic acid)	2500–3300, very broad	Often overlaps with C–H
C=O (carbonyl)	1680–1750, strong	Sharp, intense peak
C=C (alkene)	~1620–1680	Variable, weaker than C=O
N–H (amine)	3300–3500, sharp, weak	May appear as a single or double peak
C–H (alkane)	2850–2960	Often multiple peaks (sp^3 C–H stretching)

C–H (aromatic or alkene)	~3000–3100	Slightly above alkane range
C–O (ether/alcohol/ester)	1000–1300	Strong peak in fingerprint region
C–X (halogenoalkane)	500–800	Seen only in fingerprint region

Note: the data booklet will be given in the exam, you don't need to memorise anything here, this is used as an example only

E. Application: Deducing Functional Groups

Example 1:

Peak at 1700 cm^{-1} → Strong, sharp = C=O group

No peak at $2500\text{--}3300\text{ cm}^{-1}$ → No –OH = Likely a ketone

“The IR spectrum shows a **sharp absorption near 1700 cm^{-1}** , indicating the presence of a **C=O group**.

The absence of a broad O–H peak around $3200\text{--}3500\text{ cm}^{-1}$ suggests the compound is likely a **ketone**, not a carboxylic acid or alcohol.”

Example 2:

Broad peak at $\sim 3300\text{ cm}^{-1}$ + no C=O → Likely alcohol

Example 3:

Broad $2500\text{--}3300$ + sharp 1700 → Likely carboxylic acid